MSG No. 148.18 $R\bar{3}1'$ [Type II, trigonal]

Table 1: Wyckoff site: `3a`, site symmetry: `-3..1'`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4,5,6,19,20,21,22,23,24]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{3}\right]$	`[7,8,9,10,11,12,25,26,27,28,29,30]`
3	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{2}{3}\right]$	`[13,14,15,16,17,18,31,32,33,34,35,36]`

Table 2: Wyckoff site: `3b`, site symmetry: `-3..1'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,2,3,4,5,6,19,20,21,22,23,24]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{5}{6}\right]$	`[7,8,9,10,11,12,25,26,27,28,29,30]`
3	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{6}\right]$	`[13,14,15,16,17,18,31,32,33,34,35,36]`

Table 3: Wyckoff site: `6c`, site symmetry: `3..1'`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,19,20,21]`
2	$[0,\ 0,\ -z]$	`[4,5,6,22,23,24]`
3	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{3}\right]$	`[7,8,9,25,26,27]`
4	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{3}-z\right]$	`[10,11,12,28,29,30]`
5	$\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{2}{3}\right]$	`[13,14,15,31,32,33]`
6	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{2}{3}-z\right]$	`[16,17,18,34,35,36]`

Table 4: Wyckoff site: `9d`, site symmetry: `-11'`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[1,4,19,22]`
2	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,5,20,23]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,6,21,24]`
4	$\left[\frac{1}{6},\ \frac{1}{3},\ \frac{5}{6}\right]$	`[7,10,25,28]`
5	$\left[\frac{2}{3},\ \frac{5}{6},\ \frac{5}{6}\right]$	`[8,11,26,29]`
6	$\left[\frac{1}{6},\ \frac{5}{6},\ \frac{5}{6}\right]$	`[9,12,27,30]`
7	$\left[\frac{5}{6},\ \frac{2}{3},\ \frac{1}{6}\right]$	`[13,16,31,34]`
8	$\left[\frac{1}{3},\ \frac{1}{6},\ \frac{1}{6}\right]$	`[14,17,32,35]`
9	$\left[\frac{5}{6},\ \frac{1}{6},\ \frac{1}{6}\right]$	`[15,18,33,36]`

Table 5: Wyckoff site: `9e`, site symmetry: `-11'`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[1,4,19,22]`
2	$\left[0,\ \frac{1}{2},\ 0\right]$	`[2,5,20,23]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[3,6,21,24]`

continued ...

Table 5

No.	position	mapping
4	$\left[\frac{1}{6}, \frac{1}{3}, \frac{1}{3}\right]$	`[7,10,25,28]`
5	$\left[\frac{2}{3}, \frac{5}{6}, \frac{1}{3}\right]$	`[8,11,26,29]`
6	$\left[\frac{1}{6}, \frac{5}{6}, \frac{1}{3}\right]$	`[9,12,27,30]`
7	$\left[\frac{5}{6}, \frac{2}{3}, \frac{2}{3}\right]$	`[13,16,31,34]`
8	$\left[\frac{1}{3}, \frac{1}{6}, \frac{2}{3}\right]$	`[14,17,32,35]`
9	$\left[\frac{5}{6}, \frac{1}{6}, \frac{2}{3}\right]$	`[15,18,33,36]`

Table 6: Wyckoff site: `18f`, site symmetry: `11'`

No.	position	mapping
1	$[x, y, z]$	`[1,19]`
2	$[-y, x-y, z]$	`[2,20]`
3	$[-x+y, -x, z]$	`[3,21]`
4	$[-x, -y, -z]$	`[4,22]`
5	$[y, -x+y, -z]$	`[5,23]`
6	$[x-y, x, -z]$	`[6,24]`
7	$\left[x+\frac{2}{3}, y+\frac{1}{3}, z+\frac{1}{3}\right]$	`[7,25]`
8	$\left[\frac{2}{3}-y, x-y+\frac{1}{3}, z+\frac{1}{3}\right]$	`[8,26]`
9	$\left[-x+y+\frac{2}{3}, \frac{1}{3}-x, z+\frac{1}{3}\right]$	`[9,27]`
10	$\left[\frac{2}{3}-x, \frac{1}{3}-y, \frac{1}{3}-z\right]$	`[10,28]`
11	$\left[y+\frac{2}{3}, -x+y+\frac{1}{3}, \frac{1}{3}-z\right]$	`[11,29]`
12	$\left[x-y+\frac{2}{3}, x+\frac{1}{3}, \frac{1}{3}-z\right]$	`[12,30]`
13	$\left[x+\frac{1}{3}, y+\frac{2}{3}, z+\frac{2}{3}\right]$	`[13,31]`
14	$\left[\frac{1}{3}-y, x-y+\frac{2}{3}, z+\frac{2}{3}\right]$	`[14,32]`
15	$\left[-x+y+\frac{1}{3}, \frac{2}{3}-x, z+\frac{2}{3}\right]$	`[15,33]`
16	$\left[\frac{1}{3}-x, \frac{2}{3}-y, \frac{2}{3}-z\right]$	`[16,34]`
17	$\left[y+\frac{1}{3}, -x+y+\frac{2}{3}, \frac{2}{3}-z\right]$	`[17,35]`
18	$\left[x-y+\frac{1}{3}, x+\frac{2}{3}, \frac{2}{3}-z\right]$	`[18,36]`